

# Edge Geography

## is XNLP-hard for Pathwidth

## and in XP for Tree-Partition Width

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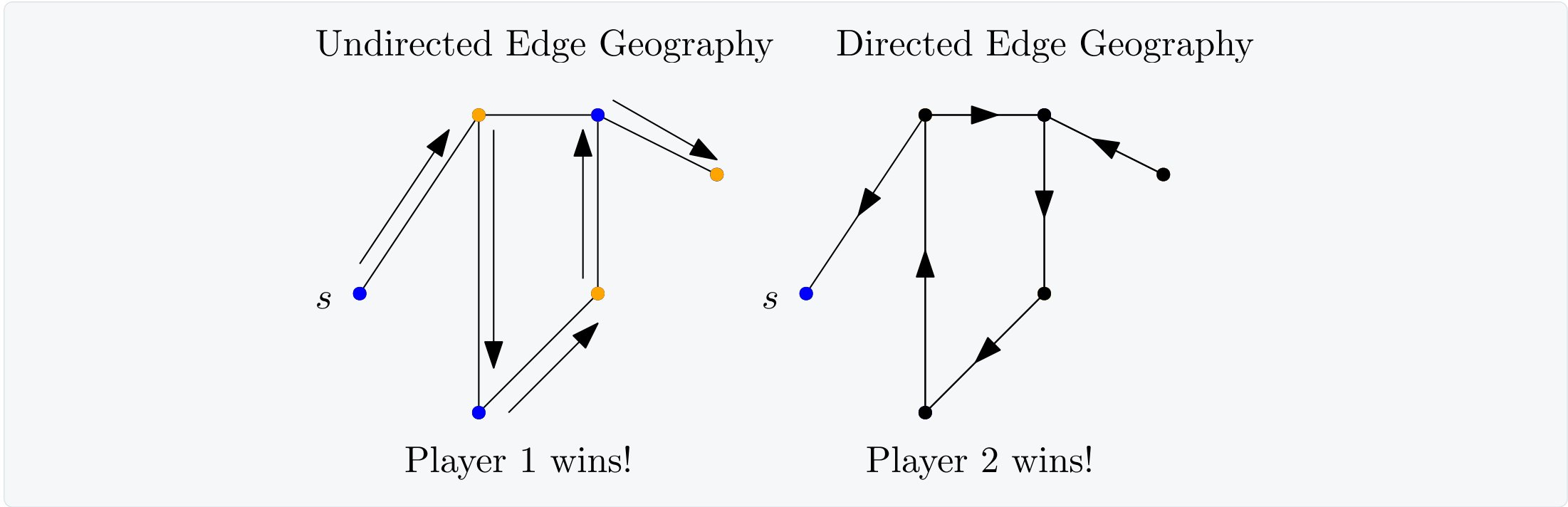
### 01 Spillet

To spillere flytter en brikke på en graf (nettverk) til en spiller ikke har noen trekk. Da vinner motstanderen!

#### Can Player 1 force a win?

Spillet er "vanskelig" generellt. Vi spør om spillet kan gjøres enkelt hvis grafen (nettverket) er "smalt"!

### 02 Hva kan fi få til?



PATHWIDTH/STIBREDDE

XNLP-hard

Directed and undirected  
Edge Geography.

Also: Directed Edge Geography is FPT for treewidth + maximum degree, with time  $f(k, \Delta) n^{O(1)}$ .

### 03 Why pathwidth is hard

**Source problem.** In *Chosen Maximum Outdegree*, orient weighted edges so that each vertex respects its outgoing-weight bound.

**Choose.** Player 1 encodes an orientation by choosing a side of each edge gadget.

**Challenge.** Player 2 tests whether a vertex exceeds its bound. A feasible orientation exists exactly when Player 1 can force a win.



The construction bounds the new pathwidth by a function of the source width. A constant-size gadget then transfers the result to undirected graphs.

**Pathwidth:** overlapping vertex bags arranged in a path; width = largest bag size minus one.

OPEN QUESTION

Is either variant in XP when parameterized by pathwidth?

XNLP-hardness rules out an FPT algorithm unless  $FPT = XNLP$ ; it does not rule out XP.

TREE-PARTITION WIDTH/TREPARTISJONSBREDDE

XP

Both variants on simple graphs.  
Running time  $n^{f(k)}$ .

### 04 Why tree partitions help

Partition the vertices into bags arranged in a tree. Edges stay inside a bag or join adjacent bags. Each bag has at most  $k$  vertices.

On simple graphs, a parent–child cut has at most  $k^2$  edges: the **ports** through which play crosses the boundary.



**Summarize.** An interface type records entry and return behavior, the player to move, and the remaining type after an excursion.

**Count.** Children of the same type are interchangeable. Store their multiplicities and process the bags bottom-up.

For fixed  $k$ , the type universe is finite. The compressed local games yield a total running time of  $n^{f(k)}$ .